



Financial Reconciliation and Monthly Closing Task

Objective: Create a financial reconciliation process, tracking key performance indicators (KPIs), and building a financial insights dashboard.

Data Sources:

1. **olist_orders_dataset.csv** - Order Details
2. **olist_order_payments_dataset.csv** - Payment Records
3. **olist_order_items_dataset.csv** - Items Ordered (including product prices)

Note: [Data Source Link](#)

Task Overview:

Create a **Monthly Financial Reconciliation Dashboard** that includes:

A) Data Preparation:

- > Load and merge the datasets on `order_id`.
 - > Create a unified dataset with relevant columns:
 - Order Date (`order_purchase_timestamp`)
 - Payment Details (`payment_value`, `payment_type`)
 - Order Status (`order_status`)
 - Product Prices (`price`, `freight_value`)
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B) Reconciliation Criteria:

1. Sales Performance:

- Total Revenue = `price + freight_value` from successfully delivered orders.
- Expected Revenue = Total of all approved orders.
- Canceled Orders = Orders with `order_status == "canceled"`.
- Late Deliveries = Orders where `order_delivered_customer_date > order_estimated_delivery_date`.

2. Payment Accuracy:

- Correct Payments = Total payments matched with the correct expected revenue.
- Under/Over Payments = Differences between expected and actual payments received.

3. Order Delays:

- Calculate the average delay in days for late orders.



- Highlight months with the highest delivery delays.
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C) Monthly Financial Dashboard Requirements:

1. Financial Overview:

- Total Revenue (Delivered Orders)
- Total Payments Received
- Expected Revenue (Approved Orders)
- % Revenue Reconciliation (Actual vs. Expected)

2. Order Breakdown:

- Total Orders by Month
- % of Canceled, Pending, and Delivered Orders
- Average Order Value

3. Delivery Insights:

- Count of Late Deliveries by Month
- Average Delay in Days

4. Interactive Features:

- Filters by Order Status, Payment Type, and Date Range.
 - Drill-Down to Order-Level Details.
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D) Bonus Analytical Tasks:

1. Profitability Analysis:

- $\text{Net Profit} = \text{price} - \text{freight_value}$ (if available).
- $\text{Gross Margin \%} = (\text{Net Profit} / \text{Total Revenue}) * 100$

2. Revenue Forecasting:

- Use previous monthly revenue trends to predict future sales and revenue growth.